Buildfest Project

Full system plan: architecture, staff experience, admin controls, connectors, and build sequence.

1. Executive summary

Buildfest Project is a private, multi agent system that lets any staff member describe what they need in plain language and receive a finished, on brand output in minutes instead of hours. It solves three problems, in order of priority.

- Data leakage. The system runs as a standalone tenant, isolated to the company. It can be demoed on a shared cloud account but is built to be redeployed inside a company's own environment.
- Wasted time and tokens. Staff no longer write and rewrite prompts by hand. A guided interview builds the technical prompt for them, in a single pass.
- Output quality and consistency. Every output is checked against the company's own brand kit, templates, and terminology before it is generated.

Email, drive, and calendar integration are a secondary, optional layer that improves day to day workflow, not the reason the system exists.

2. System architecture

2.1 Agents

- Interviewer agent. Runs on the fast, low cost tier (Claude Haiku class). Holds the conversation, asks only what is not already known, keeps latency low, and handles the general back and forth of chatting, gathering information, and structuring the brief.
- Output agent. Runs on a model tier chosen dynamically for the task at hand, see 2.3 below. Takes the finished brief, classifies what kind of output is needed, structures the payload, and either calls a connector or generates the asset directly.
- RAG knowledge base. Holds the company's brand guidelines, templates, letterheads, terminology, and asset library. Every answer the interviewer gives, and every output the system produces, is checked against this index. If something is missing, the system says so instead of guessing.

2.2 Output intent classification, and the design engine decision

Revised decision: Canva is not hardcoded at the developer level. It is a conditional, just in time connector, the same as Drive or Gmail, connected per staff member only when their specific task needs it. Reasoning: most staff work in text, documents, or spreadsheets, not visual design, so forcing a company wide Canva relationship would create cost and permission overhead for people who never touch it. The output agent instead classifies intent first, then decides whether a connector is even needed.

Scenario A: non design output (report, letter, spreadsheet)

- Triggered by words like report, memo, letter, spreadsheet, analysis.
- The output agent generates the document or spreadsheet directly, no external connector involved.
- Company letterhead, branding, and standard templates are pulled from the RAG backend automatically.
- Fast execution, no credentials requested from the staff member at all.

Scenario B: design output (deck, poster, flyer, social graphic)

- Triggered by words like presentation, slide deck, poster, flyer, social graphic.
- If the staff member already has Canva connected, the output agent proceeds straight to generating or autofilling the design.
- If Canva is not connected yet, the system shows a direct prompt: connect a Canva account, or sign up for a free one, right at the point the design is needed, not during onboarding.
- Once connected, the system resumes the exact request automatically, no need to restate anything.

This is a just in time connection flow. It removes onboarding friction for staff who only ever need text or spreadsheet output, while keeping design capable staff one click away from Canva when they need it.

2.3 Model routing logic

▲ Updated in this revision

The system is not locked to a single model per agent. It reads the nature of the task the staff member is asking for and automatically switches to the model tier that fits it, so the company only pays for heavier reasoning when the task actually needs it. There are three tiers.

- **Claude Haiku** — conversation, information gathering, and interview. Used for the guided interview itself, general chatting, and structuring a raw request into a clean brief. This is the default, lowest cost tier and covers the majority of turns in any session.
- **Claude Sonnet** — analysis and data work. Used once the task moves into reviewing, summarizing, or reasoning over data or documents, for example going through a spreadsheet, summarizing a thread, or analyzing a file the staff member has shared.
- **Claude Sonnet 5** — heavy design work. The highest tier the system is allowed to switch to. Reserved for demanding design generation, for example building or autofilling a full slide deck, poster, or brand asset through the Canva connector, where output quality and layout reasoning matter most.

The output agent decides which tier a given task needs based on its classification of the request, the same classification step already described in 2.2, and switches up only as far as the task requires. A simple chat exchange never spends Sonnet 5 level cost, and a heavy design brief is never left on Haiku.

3. Connectors

3.1 How a connector actually works

- Developer level, one time. The company registers a single developer app with each provider (Google Cloud Console for Drive and Gmail, the Canva Developer Portal, the Slack API Portal). This produces one client ID and client secret per provider, not per staff member.
- Staff level, per person. A button reads Connect Google Drive or Connect Canva. The staff member clicks it, logs into their own account, and approves a specific, limited set of permissions through OAuth 2.0, the same pattern as log in with Google.
- Token handling. The provider returns an access token and a refresh token. These are stored encrypted, tied to that individual staff member, and are used only to act on their behalf.
- Nobody on the development side ever logs into a staff member's personal account or holds a shared password for them.

3.2 Which connectors matter, and which do not

- Google Drive, essential. Holds company documents, spreadsheets, and slide decks already in use.

- Gmail, essential for Google Workspace companies. Most enterprise email on a custom company domain runs through Gmail even though it is not a personal @gmail.com address, so the Gmail API covers it directly. Microsoft 365 companies would need the Microsoft Graph API instead, planned as a phase 2 addition once the Gmail path is proven.
- Canva, essential, conditional. Covered above, connected just in time rather than during onboarding.
- Slack, useful, phase 2. Gmail covers formal, asynchronous threads. Slack covers real time team chat and operational channels. The two are complementary, not a replacement for each other, and Slack can be added once the core flow is stable.
- Notion, optional, phase 2. Common in startups and product teams, secondary to Drive, Gmail, and Canva for the first working version.
- iCloud, skipped. Built for personal Apple devices, not something enterprises should be storing company data in.

3.3 Priority connector set for the working demo

- Google Drive API, for documents, spreadsheets, and slide decks.
- Gmail API, for reading relevant email threads on request.
- Canva API, connected just in time for design specific requests.

3.4 Canva cost model, confirmed

- Building on Canva's developer platform costs the developer nothing. There is no charge for API access or for triggering design generation as a developer.
- Any subscription cost sits with the end user. A staff member with a free Canva account can still use free templates and elements. A staff member with a Canva Pro or Enterprise account unlocks their own brand kit, team assets, and premium templates inside the same flow.
- This means the company pays nothing extra for Canva access through this system beyond whatever individual staff already pay for their own Canva plan, if anything.

3.5 Canva connection flow, step by step

- Staff member clicks Connect Canva inside the app.
- A secure Canva authorization screen opens, the staff member logs in if needed and approves the requested scopes, such as reading designs, creating designs, or reading their brand kit.
- Canva redirects back with an authorization code, which the backend exchanges for an access token and a refresh token.
- From that point, the output agent calls the Canva REST API directly on the staff member's behalf to read, generate, or autofill designs.

4. File handling and uploads

- Direct file uploads inside the chat are capped at 1 megabyte. This keeps the chat itself fast and cheap to run, and avoids passing large raw files through the model context.
- For anything larger, the recommended path is to upload the file to the staff member's connected Google Drive first, then point the assistant at that file. The assistant reads it from Drive rather than carrying the full file through the chat.
- Every file that is processed, whether a small direct upload or a Drive reference, is captioned and cached once. Later references to the same file reuse that cached caption instead of reprocessing it, which directly

supports the token saving goal.

5. Access and authentication

- No public sign up. A staff account can only be created after an admin sends an invite, or after an admin approves an access request raised from the login screen (see below).
- Login is by email and password, company email or any email the admin chooses to invite. Google OAuth is not required for account login itself, it is only used later for connectors like Drive and Gmail.
- When the admin sends an invite, they include the staff member's name and job role, which is what allows the chat to greet a staff member by name and role context from their very first session.

▲ *Updated in this revision*

- Unrecognized email at login. If someone tries to log in or register with an email that has no account on file, the system does not simply refuse silently. It tells them no account was found for that email, and offers to send a request for access to the admin instead of a dead end.
- Request for access flow. The person can submit their name and email as an access request. This request lands with the admin the same way an invite would be managed, from the admin dashboard's access control area (7.1), where the admin can approve it, which converts it into a normal invited account, or decline it. Nobody gains access purely by requesting it.
- Anyone who tries to log in without an invite on file, and without an approved access request, is refused access.

6. Staff chat experience

- An LLM style chat interface. Before any conversation starts, a large centered greeting appears, personalized by name and time of day, and animates in rather than appearing instantly.
- Once a message is sent, the interface shifts into a normal scrolling conversation, with plain response text rather than heavy chat bubble styling for the assistant side.
- Workflow tools (Drive, Gmail, Canva, Slack) live under the plus / attach control in the message box, not a dedicated connectors panel — see section 12.2.
- A hamburger menu holds new chat, the library of previously generated files (section 12.4), and log out. A profile control shows the staff member's name, role, and their remaining usage credits against whatever limit the admin has set (see 7.2).
- Dark mode is true black, not navy. Light mode stays on the existing light palette. A device option follows the operating system setting and updates live if that setting changes.
- File captioning is cached, so a file that has already been read and summarized is not reprocessed on every later reference to it.

7. Admin dashboard

7.1 Access control

- Invite staff by email, name, and role.

▲ *Updated in this revision*

- Review and approve or decline access requests raised from the login screen by people without an existing account (see section 5).

- Revoke access at any time.

7.2 Usage oversight

▲ Revised: credits, not raw output count

Usage is no longer capped by counting outputs or by setting a flat daily token number. It runs on a usage credits system instead.

- A company wide default usage credit allowance applies to any staff member who does not have a custom limit set. This default is configured once and applies automatically to every new invite.
- While inviting a staff member, or at any point afterward, the admin can set that individual's own custom usage credit limit, either below the default for lighter use, or above it for heavier use.
- If the admin does not set a custom limit for a staff member, that person's usage simply falls back to the company default, no manual step required.
- View a history of what each staff member worked on and when, and how many credits each piece of work drew down.

7.3 Company knowledge and tooling

- Upload and update brand guidelines, templates, letterheads, and past work into the RAG index.
- See which connectors are active company wide versus available on demand.
- No Canva subscription to manage centrally, since it is a per staff connector, not a company wide one.

8. Step by step data flow

- Staff opens the chat and either picks a mode or types freely what they want to build.
- The interviewer agent, running on Haiku, asks only what it does not already know from the company profile.
- Once the brief is complete, the interviewer agent hands off a structured prompt to the output agent.
- The output agent classifies the request, and with it the model tier the task needs (see 2.3). Non design output is generated directly using company templates, typically on Sonnet for anything involving analysis or on Haiku for simple structured output. Design output checks for an existing Canva connection, prompts to connect if missing, then resumes automatically once connected, running on Sonnet 5.
- The staff member receives a preview and a download link, inside the same chat, and the finished file is also written into the staff member's Library and their connected Drive — see section 12.4.
- The credits drawn down by the request are deducted from that staff member's usage balance for the period, against either their custom limit or the company default (see 7.2).

9. Technical stack

Layer	Choice	Purpose
Auth and database	Supabase	Staff accounts, invites, access requests, usage credit balances, and settings
Transactional email	Resend	Sends invite emails and access request notifications on admin's behalf
Conversation model	Claude Haiku class	Fast, low cost tier for the interview step, general chat, and structuring a brief

Layer	Choice	Purpose
Analysis model	Claude Sonnet class	Mid tier for reviewing, summarizing, and reasoning over data or documents
Design model	Claude Sonnet 5 class	Highest tier the system switches to, reserved for heavy design generation through Canva
Design connector	Canva API, per staff OAuth	Produces the finished slide deck, poster, or graphic, connected just in time
File storage	Google Drive API, per staff OAuth	Large files are stored and referenced here rather than uploaded raw; generated files are also written back here for the Library
Knowledge base	RAG vector index	Company brand kit, templates, and terminology, nothing outside it
Voice input, future	Google Speech to Text	60 free minutes monthly, about 0.016 dollars per minute past that, viable at demo scale

10. File and build plan

- Chat UI, in progress. One file, the surface staff actually use. Covers the conversation, mode or skill selection, the floating nav (section 12.3), workflow tools, the Library (section 12.4), hamburger menu, profile, and theme handling.
- Admin dashboard. A separate file. Covers invites, access requests, usage history, credit limits, brand kit uploads, and connector visibility.
- Shared design tokens. Colors, type scale, and spacing values reused across both files so the two surfaces feel like one product.
- Backend logic spec. Written rules, not a screen, for RAG lookups, model routing across Haiku, Sonnet, and Sonnet 5 (section 2.3), output intent classification, the Canva just in time flow, the usage credits ledger, and the Drive based file handling path.

11. Security and privacy

- Company data stays inside the company's own isolated environment.
- Nobody can create an account without an admin invite on file, or an admin approved access request.
- The RAG layer only ever answers from the company's own uploaded material, and says so plainly when something has not been provided.
- Every connector, Drive, Gmail, or Canva, is authorized per individual staff member through OAuth, never through a shared account or a shared password.
- Large files are referenced from the staff member's own Drive rather than stored raw inside the chat system, keeping the direct upload surface small and easy to reason about.

12. Interface refinements, latest round

12.1 A named, auto switching company assistant

There is no manual mode picker. The assistant carries one configurable name per company, built as {first word of the company name} AI. The working demo is branded BuildFest AI. A company called Acme would see Acme AI instead, with no separate rebuilding step, changing the one company name value is enough to rename the assistant everywhere it appears. This single assistant runs the entire interview, reads whatever files or connected sources are relevant, writes the detailed technical prompt, and then switches itself into the right skill, PowerPoint, Word, Excel, or a deeper role such as project analyst or market analyst later on, without the staff member ever choosing a mode from a menu. The interface reflects this quietly, showing the active skill as a small label next to the assistant's name only once it has actually been detected.

▲ Updated in this revision

The assistant's app level icon and logo, the mark that represents the product itself in the browser tab, splash screen, and sidebar header, is always drawn from the company's own branding rather than a generic assistant mark. An Acme deployment shows an Acme derived icon and logo throughout; a different company's deployment shows theirs. This is distinct from the in conversation rule in 12.3, which stays unchanged, no avatar or bobblehead represents the assistant inside the message thread itself. The branding lives at the app and chrome level, not inside individual chat bubbles.

12.2 Workflow tools move under the attach control

Connectors are renamed workflow tools and relocated. They no longer live behind the hamburger menu. They live under the plus control inside the message box, alongside the upload file action. Clicking the plus control shows upload file and workflow tools, and workflow tools expands in place to reveal Canva, Gmail, Google Drive, and Slack, each shown with its own recognizable icon rather than plain text.

- Canva is now available for proactive connection here as well, so a tech comfortable staff member can connect it ahead of time if they already know they will need it.
- The just in time prompt from section 2.2 still applies as a fallback, a staff member who has not connected Canva only gets asked at the exact moment a design task needs it.
- Slack joins the list as a workflow tool for this round of interface work. The interface is being built first, the actual OAuth wiring for Slack follows the same two layer pattern described in section 3.1 once this screen is locked.
- Notion is intentionally left out for now and can be added later if a specific company engagement needs it.

12.3 Visual direction

- Glassy, premium surfaces throughout, stronger blur and softer shadows on every dropdown or panel.
- A subtle animated ambient glow, blue moving toward purple and pink, sits behind the empty state and the input area, giving the interface a sense of being alive rather than static.
- No avatar, bobblehead, or profile icon represents the assistant anywhere in the conversation. Responses read as plain flowing text on the left, matching how a modern LLM product actually looks, with the staff member's own messages as a plain neutral capsule on the right.

▲ Updated nav treatment

- Floating, glassy top nav. The header floats as a rounded, glass panel inset from the edges, close to a full pill shape, sitting on its own translucent, blurred, very dark surface with a subtle border, no drop shadow. The bar itself stays flat and dark, no gradient sits on the bar's own surface. Instead, a soft blurred gradient glow, purple moving into deep navy blue, sits just outside the bar, around its edges, as an ambient halo rather than a ring drawn on the bar, making it visually obvious the nav is floating on its own layer above the interface.
- True dark and light nav surface. In dark mode the floating nav sits on a near opaque black glass, not a lightened gray overlay, so it reads as genuinely dark rather than a translucent patch on black. In light mode it

sits on a bright, airy glass instead, keeping the same sleek, premium feel in both themes.

- Sharper, more visible glow on the voice input control. When the message box is idle, the mic icon carries a slim, sharp, slowly rotating gradient ring at a stronger visibility than a faint accent, close to the button's edge with minimal blur, so it reads as a clear, deliberate signal rather than something easy to miss. The mic stays on screen at all times, including while typing; once there is text or an attachment staged, a separate send button appears immediately to the mic's right rather than replacing it, so voice input is always one tap away.
- Rounded message input. The message box itself is a full pill, matching the rounded language of the nav and the circular mic and send controls, rather than a softly rounded rectangle.
- A more premium sidebar toggle. The menu control in the nav uses a plain, recognizable three line hamburger glyph inside a bordered square button, in keeping with the glassy, premium direction of the rest of the interface.

12.4 Library, previously generated files

A Library sits in the sidebar, above the recent chat list, showing every file the assistant has generated for that staff member. Each entry shows a skill icon (PowerPoint, Word, Excel), the output's title, and a sync status.

- Every generated file is cached locally the moment it is produced, then written automatically to the staff member's connected Google Drive in the background, the same per staff OAuth connection already used for file handling in section 4.
- While the Drive write is in flight, the Library entry shows a brief saving state. Once complete, it shows a synced indicator and the date, and becomes a direct link back into that file's own copy in Drive, so Drive stays the single source of truth rather than the chat holding a second, divergent copy.
- Because Drive is a connector the staff member has already authorized (section 3.1), no separate storage system or additional permission prompt is needed to support the Library; it reuses the existing Drive connection and the file captioning and caching behavior already described in section 4.

12.5 Small experience refinements

- Empty states with direction. The recent chat list no longer shows an inert "No previous chats yet" line. It reads "Nothing here yet — start your first output above" and is itself a button that focuses the input, turning empty space into an invitation to act rather than a dead end.
- Slash quick starts. Typing / in the message box, at the very start of a conversation, surfaces a short list of common starting points, such as New pitch deck or Q3 report. Picking one seeds that exact request and sends it immediately, skipping straight into the interview rather than making the staff member type it out or pass through small talk first.
- A noticed skill switch. The first moment the assistant detects which skill a request needs, the small "PowerPoint mode" style badge next to the assistant's name plays a brief, one time flash and settle animation as it appears, rather than a plain fade in. This makes the automatic switch legible as something that just happened, without adding a persistent visual element or interrupting the flow.
- Library search. Once a staff member's Library holds more than a handful of files, a small search field appears at the top of the panel so they can filter by title instead of scrolling. Below that threshold the field stays hidden, since a short list is faster to scan than to search.